B2B SaaS Customer Retention Analysis

End-to-End Business Analysis, Data Cleaning & Power BI Dashboard Project

Role: Business Analyst
Scope: 50,000 Global Accounts
Annual Run-Rate: $399.2M ARR

1. Project Overview & The Business Problem

CloudScale is a B2B subscription software company serving **50,000 business accounts** across 8 countries (USA, Canada, UK, Germany, France, Singapore, Australia, and India). The company generates **$33.27 Million in Monthly Recurring Revenue (MRR)**, which totals **$399.2 Million Annual Recurring Revenue (ARR)**.

The Problem: The company was losing **$4.75 Million in revenue every year** due to customer cancellations (916 accounts left). Leadership lacked visibility into why customers were leaving, and Account Managers had no early warning signs to reach out to customers before they cancelled.

$399.2M	$4.75M	1.83%	+$1.80M
TOTAL ANNUAL REVENUE (ARR)	ANNUAL LOST REVENUE (CHURN)	CUSTOMER CHURN RATE	RECOVERABLE ANNUAL REVENUE

My Role as Business Analyst: Analyze the 50,000 raw customer records, clean up data errors, identify the true drivers of customer cancellations, build an interactive 3-page Power BI dashboard, and provide management with actionable recommendations to recover lost revenue.

2. Data Audit & How I Cleaned It

The raw dataset came with 50,000 rows and 18 columns. During the data quality audit, I found and addressed three main items:

- **Fixed Date Logic (59 rows):** 59 records had a `Last_Active_Date` that was recorded *earlier* than the customer's `Signup_Date` (a database logging flaw). I updated the script to set the last active date equal to the signup date so customer tenure wouldn't calculate as a negative number.
- **Cleaned Metric Formatting:** The `Feature_Adoption_Rate` column was saved as text with % symbols (e.g., `"44%"`). I converted these into clean decimal numbers so Power BI could calculate mathematical averages.
- **Star Schema Modeling:** Rather than leaving 50,000 rows in one large, slow flat file, I organized the data into a proper relational **Star Schema**:
  - **1 Central Fact Table:** `Fact_Subscriptions` (contains customer revenue, seats, tickets, and tenure).
  - **5 Dimension Tables:** `Dim_Customer`, `Dim_Plan`, `Dim_AccountManager`, `Dim_Geography`, and `Dim_Date`.

3. What the Data Actually Revealed (The 4 Big Findings)

Comparing customers who renewed against those who cancelled revealed four primary drivers:

Driver	Retained Customers	Cancelled Customers	Business Impact
1. Failed Payments	1.14% churn rate (Active)	**14.49% churn rate (Failed)**	**12.7x higher churn!** Failed credit cards and expired billing contacts caused 32% of all company churn ($1.54M ARR).
2. Seat Underutilization	64.2% average seat usage	**47.0% average seat usage**	Customers who use less than 50% of purchased licenses cancel at renewal because CFOs view the unused seats as waste.
3. Feature Adoption Gap	70.1% platform adoption	**52.1% platform adoption**	Customers who fail to adopt core features see lower product value, making it easier for them to cancel.
4. Support Ticket Spikes	3.3 tickets per month	**5.0 tickets per month**	When tickets cross 4 in a month, churn risk rises sharply due to product

Driver	Retained Customers	Cancelled Customers	Business Impact
			friction and unresolved bugs.

4. The Power BI Dashboard Solution

I built an executive 3-page Power BI dashboard designed for distinct user roles:

- **Page 1: Executive Overview (For VP of Customer Success & CRO):** Displays headline KPI cards (ARR, Churn Rate, Active Accounts), a revenue breakdown by Plan Tier (showing Starter plans carry 75% of churn), a payment health donut chart, and revenue across the 8 global countries.
- **Page 2: Customer Health & Early Warnings (For RevOps):** Features a 2D quadrant matrix plotting Feature Adoption against Seat Utilization to categorize accounts into Champions, Under-Adopters, and High-Risk clients. Also includes ticket volume correlations and health distributions.
- **Page 3: Account Manager Calling Sheet (For Customer Success Managers):** A manager scorecard comparing all 5 Account Managers (Aisha Khan had the best retention at 1.63% churn; Sarah Jenkins had the highest churn at 1.96%), paired with an interactive, searchable calling sheet with 1-click CSV export so CSMs can immediately contact at-risk accounts.

5. Strategic Recommendations & Financial Impact

To address the $4.75 Million in annual cancellations, I presented three straightforward business initiatives:

Initiative	Action Plan	Projected Benefit
1. Automated Payment Recovery	Implement automated email/SMS reminders when credit card payments fail (at 24h, 72h, and 7 days) with a self-service payment update link.	**+$1,001,000 ARR Saved** (65% cut in failed payment churn)
2. Low-Seat-Usage CSM Playbook	Create automated dashboard alerts when seat utilization drops below 50%, prompting Account Managers to offer user training 90 days before renewal.	**+$495,000 ARR Saved** (Rescuing at-risk SMB accounts)
3. Support SWAT Escalation	Automatically route customer accounts with 5 or more open tickets to a Senior Solutions Engineer before the customer becomes frustrated enough to quit.	**+$305,000 ARR Saved** (Preventing friction-based churn)
TOTAL PROJECTED ANNUAL BENEFIT		**+$1,801,000 ARR Saved**

> **Bottom-Line Result:** Implementing these three initiatives recovers **$1.80 Million per year**, reducing the company's annual churn leakage by **37.9%**.

6. Interview Summary (How to Describe This Project)

When discussing this project in an interview, here is the clear, structured story:

- **Situation:** CloudScale, an enterprise SaaS business with $399M in ARR, was losing $4.75M annually to churn without understanding the underlying causes.
- **Task:** As Business Analyst, I owned requirements gathering, data cleansing in Python, Star Schema data modeling, DAX metric creation, Power BI dashboard design, and financial ROI recommendations.
- **Action:** I cleaned 50,000 customer records, resolved 59 date errors, and built a relational model. My analysis uncovered that payment failures caused 32% of churn and that accounts using less than 50% of their seats cancelled at renewal. I created a 3-page Power BI dashboard with early warning quadrants and customer calling sheets.
- **Result:** I provided 3 concrete recommendations (automated payment recovery, seat utilization alerts, and support escalation) that project a **$1.8 Million annual revenue recovery** (38% churn reduction).